



FOR JAVA DEVELOPERS MOVING TO NESTJS

100 payments. 11 traces.

Spring Boot to NestJS, episode 37: OpenTelemetry

A trace is usually a decision made at the root and propagated downstream.

Spring Boot's OpenTelemetry starter samples 10% of root requests by default: 9 of 100 in our run. The Node SDK sampled all 100.

A RestClient built from Spring's injected builder carried the trace into the next service. RestClient.create() did not.

100 payments through Spring at its default, each calling a Nest ledger that samples everything: 11 ledger traces. Nest followed the parent.

In scope

- The Spring Boot 4 starter and the Node SDK
- Sampling defaults
- Propagation across HTTP
- The parent's decision

Not in scope

- Metrics and logs over OTLP
- Collector pipelines and tail sampling

At the time of recording management.tracing.sampling.probability defaults to 0.1.

01 The measurements

SPRING, 100 ROOTS, DEFAULTS	9 traces
NEST, 100 ROOTS, DEFAULTS	100 traces
RestClient FROM THE BUILDER	one trace, both services
RestClient.create()	Spring only
NEST fetch INTO SPRING	one trace, both services
100 VIA SPRING DEFAULT, NEST LEDGER	11 ledger traces
100 VIA SPRING AT 1.0	100 ledger traces

Traces counted in Jaeger through its query API.

02 Spring: sample on purpose, build clients from the builder

Set the root sampling probability, and build every HTTP client from the injected `RestClient.Builder` so it is instrumented.

```
application.properties
```

```
management.tracing.sampling.probability=1.0
management.opentelemetry.tracing.export.otlp.endpoint=http://collector:4318/v1/traces
```

```
PaymentsController.java
```

```
PaymentsController(RestClient.Builder builder) {
    this.ledger = builder.baseUrl(ledgerUrl).build(); // instrumented
    // RestClient.create(ledgerUrl) is not
}
```

builder: one trace, both services; create(): Spring only

Run on a real machine, 2026-10-01, by `scripts/verify-otel.sh` in the course repository.

03 Nest: the Node SDK, configured by environment

Start the SDK before the app, configured by the standard `OTEL_` variables. Roots are all sampled by default; a sampled parent is followed.

```
shell
```

```
OTEL_SERVICE_NAME=ledger \
OTEL_EXPORTER_OTLP_ENDPOINT=http://collector:4318 \
OTEL_TRACES_SAMPLER=parentbased_traceidratio OTEL_TRACES_SAMPLER_ARG=0.25 \
node --import @opentelemetry/auto-instrumentations-node/register dist/main.js
```

roots: 100 of 100; behind a 10% Spring root: 11 of 100

Run on a real machine, 2026-10-01, by `scripts/verify-otel.sh` in the course repository.

Before you trust a trace

1. Which service is the root, and what does it sample?
2. Do downstream services follow the parent?
3. Is every HTTP client on the instrumented path?
4. Does one request show every service in one trace?
5. Is the sampling rate a decision someone made?

Everything else

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